

Stage 1 — identity + change detection
THE BOTTLENECK

100 units → 97.9 ms

1,000 units → 10.48 s

10,000 units → MemoryError

→ dirty set →

Stage 2 — selective recompile
given a dirty set

1,000 units → 3.62× vs full
rebuild

10,000 units → 4.31× vs full
rebuild

~1% of artifacts rebuilt